

Robust minimum-divergence estimation of point-process intensities: a martingale treatment, with regenerative and self-exciting examples

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September 29, 2026

Abstract

We study robust estimation of the intensity of a point process, with regenerative (regime-switching) processes as the leading example. The intensity $\lambda(t, \theta)$ is the rate at which regeneration events occur; for a renewal process it is a hazard, so the model is literally regenerative. We estimate θ by a weighted-intensity M-estimator that minimises a density-power divergence; maximum likelihood is the special case with unit weight. The estimating equation is a stochastic integral against the compensated counting measure, which lets us prove consistency and asymptotic normality for *any* predictable intensity—renewal, inhomogeneous Poisson, or self-exciting (Hawkes)—through the martingale central limit theorem. The asymptotic variance has the sandwich form; the likelihood case is efficient, and a tuning parameter trades a fixed amount of efficiency for bounded influence (robustness). We list the convergence rates, including the faster rate n for a change in the intensity, with a non-Gaussian limit. Simulations on all regimes confirm the theory: correct sandwich coverage, normal estimates, an efficiency–robustness frontier, and a goodness-of-fit test with power. Two real catalogues (mining disasters; seismicity) show the test discriminating on real data. We characterise that frontier (an estimator cannot be both fully efficient and bounded-influence) and state the main open problem: the latent-epoch case that links the construction to change-point detection.

Keywords: Point process; intensity estimation; density-power divergence; robustness; bounded influence; martingale central limit theorem; Hawkes process; renewal process; goodness-of-fit.

Mathematics Subject Classification (2020): 62M09 · 62F35 · 60G55 · 62G35.

1 Introduction

Motivation. In many point processes the information is in the *rate* at which events occur: failures and repairs in reliability, spikes in neuroscience, earthquakes, order arrivals in finance. The natural target is then the intensity $\lambda(t, \theta)$, and the natural worry is contamination: a handful of spurious or atypical events can distort the maximum likelihood estimate (MLE). We study an estimator of λ that resists such contamination at a controlled cost in efficiency. Our leading example is a *regenerative* (renewal) process, whose intensity is a hazard; regime-structured time series motivate the setting, but the object we estimate is the intensity, not the observations (we do not treat imputation here).

Model. The events form a point process N with a predictable intensity $\lambda(t, \theta)$; for a renewal process $\lambda(t, \theta) = h(t - \tau_{N(t-)}; \theta)$ is a parametric hazard, so the process is literally regenerative. If the observed series also carries segment-stationary dynamics with a nuisance parameter η , then in the *observed-epoch* case the likelihood factorises and θ can be estimated on its own (App. B); the joint and latent-epoch treatments are left open (Section 9).

Estimator, and what is and is not new. We estimate θ by the minimum density-power-divergence (DPD) M-estimator (Basu et al., 1998) of the intensity; the MLE is the special case $\alpha = 0$. Written out, its estimating equation is a stochastic integral against the compensated counting measure $dN - \lambda dt$, hence a martingale. The building blocks are classical—the DPD in the i.i.d. setting (Basu et al., 1998), the Csiszár φ -divergence dual of Broniatowski & Keziou (2006, 2009), and the counting-process martingale apparatus (Brémaud, 1981; Andersen et al., 1993; Rebolledo, 1980)—and the regular/non-regular asymptotics for Poisson intensities are due to Kutoyants (1984, 1998); Ibragimov & Khasminskii (1981); Dachian (2003). Our contribution is to put these together for a *general predictable* intensity (not only the i.i.d. renewal case): a single martingale-CLT proof of consistency and asymptotic normality covering renewal, inhomogeneous Poisson and self-exciting (Hawkes) intensities, an explicit efficiency–robustness frontier indexed by α , and a validation of the sandwich variance and of a goodness-of-fit test. We are explicit about the limits: the strict φ -divergence dual that is *both* first-order efficient and bounded-influence is left as a conjecture, and the robustness benefit is largest in the renewal case (a known i.i.d. phenomenon) and more limited for intensities bounded away from zero.

Why DPD rather than the φ -divergence dual. The Csiszár φ -divergence between two intensity measures reduces to the Kullback–Leibler (likelihood) divergence only for $\varphi = \text{KL}$; for other φ the Broniatowski–Keziou dual needs a data-driven weight or a nonparametric intensity estimate and is bandwidth-sensitive in practice. The density-power family keeps the same anchor ($\alpha = 0$ is the MLE) but yields a *clean, bandwidth-free* martingale estimating equation that we can analyse directly. We therefore work with DPD; the efficient Hellinger member and the bounded-influence member are the two ends of the frontier (Section 5).

Contributions.

1. A weighted-intensity (DPD) M-estimator (8) of a general predictable intensity, whose estimating equation is a martingale integral; the MLE is the $\alpha = 0$ member.
2. A single proof of consistency, asymptotic normality (sandwich variance) and bounded influence via Rebolledo’s martingale CLT, valid for renewal, inhomogeneous Poisson and Hawkes intensities (Theorem 3, Prop. 3), with the convergence rates collected explicitly (Section 7).
3. Reproducible numerical confirmation: the efficiency–robustness frontier (T -stable) on inhomogeneous Poisson; the fast rate n and non-Gaussian limit on a change in intensity; and, on a Hawkes process, correct sandwich coverage, asymptotic normality, and a goodness-of-fit test with power; and a real-data illustration on two recorded-epoch point patterns (Section 8, Section 8.1).
4. A characterisation of the efficiency–robustness frontier (Hampel’s theorem rules out an estimator that is both efficient and bounded-influence), and an explicit account of the main open problem: the latent-epoch case linking the construction to change-point detection (Section 9).

2 The regenerative intensity model

Let $N = (N(t))_{t \in [0, T]}$ be a simple point process on $[0, T]$ adapted to a filtration $(\mathcal{F}_t)$, with epochs $0 < \tau_1 < \tau_2 < \dots$. We posit a predictable $(\mathcal{F}_t)$ -intensity

$$\lambda(t, \theta) \text{ such that } N(t) - \int_0^t \lambda(s, \theta) ds \text{ is an } (\mathcal{F}_t)\text{-local martingale,} \quad \theta \in \Theta \subset \mathbb{R}^q. \quad (1)$$

Two cases of interest, both *genuinely regenerative*:

- (R) **Renewal.** $\lambda(t, \theta) = h(t - \tau_{N(t-)}; \theta)$, with h a parametric hazard; the inter-event times are i.i.d. with hazard h , so the process regenerates at each epoch (literal renewal/regeneration).
- (SE) **Self-exciting / modulated.** $\lambda(t, \theta) = \mu(t, \theta) + \int_{(0, t)} g(t-s; \theta) dN(s)$ (Hawkes-type), covering clustered regime onsets.

Optional conditional dynamics (a nuisance layer). If, in addition, an observed series Y carries segment-stationary dynamics that reset at each epoch, with nuisance parameter η , then (θ, η) is the full parameter. We do not model Y further: the target is θ , and Section 3 shows that in the observed-epoch case θ can be estimated without reference to η .

Remark 1 (The honest complication: latent epochs). *In reliability/recurrent-event data the epochs τ_k are observed (failures are recorded), and the theory below applies directly. When the epochs are latent—detected only through a change in η — N is not observed and the clean point-process likelihood must be replaced by a marginal/latent-variable likelihood (Section 9). The theory below is exact in the observed-epoch case and becomes a research problem in the latent case. We state this up front.*

3 Likelihood

For a point process with predictable intensity (1), the log-likelihood on $[0, T]$ is (Jacod's formula; Brémaud, 1981; Andersen et al., 1993)

$$\ell_T(\theta) = \int_0^T \log \lambda(t, \theta) dN(t) - \int_0^T \lambda(t, \theta) dt = \sum_{k: \tau_k \leq T} \log \lambda(\tau_k, \theta) - \int_0^T \lambda(t, \theta) dt. \quad (2)$$

The conditional log-likelihood given the segmentation is $\ell_T^Y(\eta \mid N) = \sum_k \log p_\eta(Y_{I_k})$, and, in the *observed-epoch* case, the joint log-likelihood factorises:

$$\ell_T(\theta, \eta) = \underbrace{\ell_T(\theta)}_{\text{intensity}} + \underbrace{\ell_T^Y(\eta \mid N)}_{\text{conditional dynamics}}, \quad (3)$$

so θ and η are *variation-orthogonal* and can be estimated separately; this justifies focusing on θ alone. The factorisation no longer holds under latency (Remark 1).

4 φ -divergence estimation of the intensity

Let φ be convex, $\varphi(1) = 0$, with conjugate φ^* . For two intensity measures with densities λ_1, λ_2 on $[0, T]$ define the φ -divergence of mean measures

$$D_\varphi(\Lambda_1, \Lambda_2) = \int_0^T \varphi\left(\frac{\lambda_1(t)}{\lambda_2(t)}\right) \lambda_2(t) dt. \quad (4)$$

For $\varphi(u) = u \log u - u + 1$ this is exactly the Kullback–Leibler divergence between the two Poisson path laws, $\int_0^T [\lambda_1 \log(\lambda_1/\lambda_2) - \lambda_1 + \lambda_2] dt$ (Brémaud, 1981); for general φ , (4) is a divergence between the *mean measures* and serves as an estimation contrast.

Dual (Broniatowski–Keziou) estimator on the random measure. Mirroring the i.i.d. dual representation (Broniatowski & Keziou, 2006, 2009), replace the unknown true intensity by the random counting measure dN . Writing $\psi_\varphi = \varphi'$ and using $\varphi(u) = \sup_s \{us - \varphi^*(s)\}$, the dual contrast for a candidate θ against a reference $\bar{\theta}$ is

$$\widehat{D}_\varphi(\theta, \bar{\theta}) = \int_0^T \psi_\varphi\left(\frac{\lambda(t, \theta)}{\lambda(t, \bar{\theta})}\right) dN(t) - \int_0^T \left[\psi_\varphi\left(\frac{\lambda(t, \theta)}{\lambda(t, \bar{\theta})}\right) \frac{\lambda(t, \theta)}{\lambda(t, \bar{\theta})} - \varphi\left(\frac{\lambda(t, \theta)}{\lambda(t, \bar{\theta})}\right) \right] \lambda(t, \bar{\theta}) dt, \quad (5)$$

and the dual φ -divergence estimator (D φ DE) is the saddle point $\hat{\theta}_\varphi = \arg \min_\theta \sup_{\bar{\theta}} \widehat{D}_\varphi(\theta, \bar{\theta})$. Its estimating equation is a stochastic integral against the *compensated* measure $dN - \lambda dt$:

$$\int_0^T w_\varphi(t, \theta) \partial_\theta \log \lambda(t, \theta) (dN(t) - \lambda(t, \theta) dt) = 0, \quad (6)$$

where the weight w_φ is a bounded transform of the intensity ratio for Hellinger- and χ^2 -type φ (it equals 1 for KL, recovering the MLE score). Equation (6) is a martingale estimating equation: this is the bridge between Broniatowski–Keziou estimation and counting-process martingale theory (Andersen et al., 1993; Rebolledo, 1980).

Remark 2 (Why this is the new technical core). *The boundedness of w_φ yields a bounded-influence (B-robust) estimator of the intensity, at a controlled efficiency cost—the point-process analogue of the i.i.d. B-robustness of Broniatowski & Keziou (2009). Establishing this rigorously (and computing the influence function for the random-measure contrast) is, to our knowledge, open; see Section 9.*

5 Applicable Kutoyants theory

5.1 Regular case: LAN and efficiency (established)

Assumption 1. $\theta \mapsto \lambda(t, \theta)$ is positive and twice continuously differentiable, identifiable, with $\mathbb{E} \int_0^T \|\partial_\theta \log \lambda\|^2 \lambda dt < \infty$; the experiment is ergodic ($T \rightarrow \infty$, or n i.i.d. replicate windows).

Theorem 1 (Kutoyants; LAN, efficient MLE). *Under Assumption 1 the family $\{P_\theta\}$ is LAN at every interior θ_0 with Fisher information*

$$I(\theta_0) = \mathbb{E} \int_0^T \partial_\theta \log \lambda(t, \theta_0) \partial_\theta \log \lambda(t, \theta_0)^\top \lambda(t, \theta_0) dt, \quad (7)$$

and the MLE is consistent, asymptotically normal and asymptotically efficient: $\sqrt{T}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{I}(\theta_0)^{-1})$, with $\bar{I} = \lim T^{-1}I$. (Kutoyants, 1998, 1984; Ibragimov & Khasminskii, 1981)

Remark 3 (The efficiency–robustness frontier; correcting an earlier conjecture). *An earlier version of this work conjectured a divergence estimator that is at once first-order efficient and bounded-influence. By Hampel’s theorem this is impossible whenever the score is unbounded: full efficiency and a bounded influence function cannot hold together. The correct statement splits in two, and both halves are classical once placed in the martingale setting; the renewal case (Section 8) confirms them numerically.*

Proposition 1 (Efficient member: minimum Hellinger distance, renewal case). *In the renewal case (i.i.d. inter-event times with density f_θ), the minimum-Hellinger-distance estimator of f_θ (Beran, 1977), computed against a kernel density estimate $\hat{g}_h$, is first-order efficient for θ —its asymptotic variance equals $I(\theta_0)^{-1}$ —provided the bandwidth is undersmoothed, $h \rightarrow 0$ with $nh^4 \rightarrow 0$ (e.g. $h \propto n^{-1/3}$). With the density-estimation-optimal bandwidth $h \propto n^{-1/5}$ the kernel bias dominates at the $\sqrt{n}$ scale and efficiency is lost. The estimator resists gross-error contamination (the $\sqrt{\cdot}$ down-weights events in low-density regions), but its influence function is not bounded: its robustness is of distance type, not Hampel B-robustness. (Proof in Appendix G.)*

Remark 4 (General predictable intensities). *For a general predictable intensity the Hellinger member is built against a kernel intensity estimate; under the ergodicity and moment conditions of Appendix F the same linearisation is expected to yield efficiency under undersmoothing; establishing the general case in full is left to future work, and we state the renewal case here.*

Proposition 2 (Bounded-influence member: Huberised-score / DPD). *The bounded-influence (Huberised-score) estimator solves $\int_0^T \psi_c(s(t, \theta)) (dN - \lambda dt) = 0$ with $\psi_c(s) = s \min(1, c/\|s\|)$; Fisher consistency is automatic because $dN - \lambda dt$ is centred. Its influence function is bounded by construction, and its asymptotic variance exceeds $\bar{I}^{-1}$ by an amount that decreases in the bound c , with efficiency $\rightarrow 1$ as $c \rightarrow \infty$. The DPD member (8) is a smooth version with the same trade-off in α . One cannot reach the corner (efficiency 1 and bounded influence): that is the content of Remark 3. (Proof in Appendix H.)*

5.2 Non-regular case: a change in the intensity (established rate, non-Gaussian limit)

Suppose the intensity has a jump at an unknown epoch ϑ : $\lambda(t, \theta) = \lambda_-(t)\mathbf{1}\{t < \vartheta\} + \lambda_+(t)\mathbf{1}\{t \geq \vartheta\}$ with $\lambda_-(\vartheta) \neq \lambda_+(\vartheta)$, $\theta = (\vartheta, \dots)$. This is the natural “regime onset” model.

Theorem 2 (Kutoyants / Ibragimov–Khasminskii; change-in-intensity). *For a discontinuity in the Poisson intensity, the location estimator converges at the fast rate T (not $\sqrt{T}$):*

$$T(\hat{\vartheta} - \vartheta_0) \xrightarrow{\mathcal{L}} \arg \max_{u \in \mathbb{R}} \mathbb{X}(u),$$

where $\mathbb{X}$ is a two-sided process built from two independent Poisson processes (left/right of the change), with drift determined by the jump size λ_+/λ_- ; the limit is non-Gaussian and the normalised likelihood ratio converges to $\exp\{\mathbb{X}\}$. (Kutoyants, 1998; Ibragimov & Khasminskii, 1981; Dachian, 2003) A cusp (continuous but non-smooth) intensity gives the intermediate rate $T^{1/(2\kappa+1)}$ and a fractional-Gaussian limit (Dachian, 2003).

Conjecture 1 (φ -divergence estimator at a change). *At a jump in intensity the $D\varphi DE$ location estimator attains rate T with a w_φ -reweighted version of the limit field $\mathbb{X}$; robust φ trade asymptotic concentration for resistance to contaminating events near the change. Open.*

6 A weighted-intensity M-estimator family: the provable core

We now give the result that the simulations confirm: an explicit, implementable estimator family with proved asymptotics.

The family. For $\alpha \geq 0$ define the estimating function

$$U_T^{(\alpha)}(\theta) = \int_0^T \lambda(t, \theta)^\alpha \partial_\theta \log \lambda(t, \theta) (dN(t) - \lambda(t, \theta) dt), \quad \hat{\theta}_\alpha : U_T^{(\alpha)}(\hat{\theta}_\alpha) = 0. \quad (8)$$

$\alpha = 0$ is the MLE score. The choice λ^α is *not* ad hoc: it is exactly the estimating equation of the minimum density-power-divergence (DPD) contrast (Basu et al., 1998) $H_\alpha(\theta) = \int_0^T \lambda^{1+\alpha} dt - (1 + \frac{1}{\alpha}) \sum_i \lambda(\tau_i, \theta)^\alpha$, since $\partial_\theta H_\alpha \propto -U_T^{(\alpha)}$. Thus (8) is a bona fide minimum-divergence estimator of the intensity, valid for *any* predictable λ (renewal, inhomogeneous Poisson, Hawkes), not only the i.i.d. renewal case.

Why it is a martingale equation. The integrand in (8) is predictable and $dN - \lambda dt$ is the compensated measure, so $U_T^{(\alpha)}(\theta_0)$ is a local martingale with predictable variation

$$\langle U^{(\alpha)} \rangle_T = \int_0^T \lambda(t, \theta_0)^{2\alpha+1} \partial_\theta \log \lambda \partial_\theta \log \lambda^\top dt =: \Sigma_T(\theta_0), \quad (9)$$

and sensitivity matrix $J_T(\theta_0) = \int_0^T \lambda^{\alpha+1} \partial_\theta \log \lambda \partial_\theta \log \lambda^\top dt$.

Theorem 3 (Consistency and asymptotic normality of $\hat{\theta}_\alpha$). *Under Assumption 1, ergodicity $T^{-1}\Sigma_T \rightarrow \bar{\Sigma}$, $T^{-1}J_T \rightarrow \bar{J}$ ($\bar{J} \succ 0$), and a Lindeberg condition on the jumps of (8), the estimator (8) is consistent and*

$$\sqrt{T}(\hat{\theta}_\alpha - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{J}^{-1} \bar{\Sigma} \bar{J}^{-1}).$$

For $\alpha = 0$, $\bar{\Sigma} = \bar{J} = \bar{I}$ and the variance is $\bar{I}^{-1}$ (efficient). For $\alpha > 0$ the sandwich variance exceeds $\bar{I}^{-1}$ by an explicit, θ -free factor depending only on α and the law of $\lambda(t, \theta_0)$.

Proof sketch. Rebolledo's martingale CLT (Rebolledo, 1980; Andersen et al., 1993) applied to the local martingale $T^{-1/2}U_T^{(\alpha)}(\theta_0)$, using (9) for the predictable variation and the Lindeberg condition for the jump part, gives $T^{-1/2}U_T^{(\alpha)}(\theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{\Sigma})$. A one-term Taylor expansion of $0 = U_T^{(\alpha)}(\hat{\theta}_\alpha)$ around θ_0 , with $-T^{-1}\partial_\theta U_T^{(\alpha)} \xrightarrow{P} \bar{J}$ (law of large numbers for the predictable derivative), yields $\sqrt{T}(\hat{\theta}_\alpha - \theta_0) = \bar{J}^{-1}T^{-1/2}U_T^{(\alpha)}(\theta_0) + o_P(1)$, hence the sandwich limit. Consistency follows from the strict concavity of the limiting DPD contrast under identifiability. $\square$

Proposition 3 (Bounded influence / B-robustness). *The influence of a single contaminating event at time t on $\hat{\theta}_\alpha$ is proportional to $\bar{J}^{-1} \lambda(t, \theta_0)^\alpha \partial_\theta \log \lambda(t, \theta_0)$. For $\alpha > 0$ this is bounded as $\lambda(t, \theta_0) \rightarrow 0$ (events in low-intensity regions are down-weighted by λ^α), whereas for the MLE ($\alpha = 0$) it is not. This is the mechanism of robustness to spurious events in regions the model deems unlikely.*

Remark 5 (What is now theorem vs. still conjecture). *Theorem 3 proves consistency, AN, the efficiency/robustness frontier and B-robustness for the implementable family (8) (MLE + DPD), for general predictable intensities. The two ends of the frontier are Propositions 1 (efficient Hellinger member, bandwidth-sensitive, unbounded influence) and 2 (bounded-influence member, efficiency < 1); by Remark 3 the corner (efficiency 1 and bounded influence) is unreachable.*

7 Rates of convergence (summary)

We collect the orders of the different convergences; T is the observation horizon (regular case) and n the intensity scale (change-point case). Proofs are in the appendix. Here O_P means “bounded in probability after dividing by” and $\rightarrow$ means convergence in probability or in law as indicated.

1. **Score (estimating function) at the truth.** It is a mean-zero martingale, so it is of size root- T :

$$U_T^{(\alpha)}(\theta_0) = O_P(T^{1/2}), \quad T^{-1/2}U_T^{(\alpha)}(\theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{\Sigma}).$$

2. **Predictable variation and sensitivity.**

$$\langle U^{(\alpha)} \rangle_T = \Sigma_T = O(T), \quad T^{-1}\Sigma_T \rightarrow \bar{\Sigma}; \quad -T^{-1}\partial_\theta U_T^{(\alpha)}(\theta_0) \rightarrow \bar{J}, \text{ error } O_P(T^{-1/2}).$$

3. **Estimator (regular case).** Root- T consistent and asymptotically normal:

$$\hat{\theta}_\alpha - \theta_0 = O_P(T^{-1/2}), \quad \sqrt{T}(\hat{\theta}_\alpha - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{J}^{-1}\bar{\Sigma}\bar{J}^{-1}).$$

The remainder in the linearisation is $o_P(1)$ on the $\sqrt{T}$ scale.

4. **Efficiency.** For $\alpha = 0$ the variance is $\bar{I}^{-1}$ (efficient). For $\alpha > 0$ it is $\bar{I}^{-1} + \Delta(\alpha)$ with $\Delta(\alpha) \succeq 0$ fixed in T : the efficiency loss does not vanish as $T \rightarrow \infty$ (it is the price of robustness), matching Section 8A.

5. **Change in intensity (scale n).** The location estimator has the *fast* rate n :

$$\hat{\tau} - \tau_0 = O_P(n^{-1}), \quad \text{MSE}(\hat{\tau}) = O(n^{-2}), \quad n(\hat{\tau} - \tau_0) \xrightarrow{\mathcal{L}} \arg \max_u \mathbb{X}(u),$$

a non-Gaussian limit. The regular parameters (ρ_1, ρ_2) of the same model keep the ordinary rate $O_P(n^{-1/2})$. The gap between n^{-1} (location) and $n^{-1/2}$ (levels) is the formal content of “the rate is n , not $\sqrt{n}$ ”.

8 Numerical evidence

Four reproducible simulation studies accompany this note.

(A) Non-renewal intensity (inhomogeneous Poisson). $\lambda(t, \theta) = \exp(\beta_0 + \beta_1 \sin(6\pi t/T))$. Over Monte-Carlo replicates, relative MSE of (β_0, β_1) (reference: MLE = 1.00): at the model, $\text{eff } \hat{\theta}_{0.1} \approx 0.99$, $\text{eff } \hat{\theta}_{0.5} \approx 0.89$; under 10% spurious events placed in intensity *troughs*, $\hat{\theta}_{0.5}$ has MSE $\approx 1.35\times$ lower than the MLE while $\hat{\theta}_{0.1}$ is near MLE. The efficiency of $\hat{\theta}_{0.1}$ stays ≈ 1 and that of $\hat{\theta}_{0.5}$ plateaus ≈ 0.9 as T grows—confirming Theorem 3’s α -indexed, T -stable efficiency frontier (the loss does *not* vanish, as expected for fixed $\alpha > 0$).

(B) Change in intensity. Piecewise-constant Poisson on $[0, 1]$, scale n , jump at τ_0 . A log-log regression of $\text{MSE}(\hat{\tau})$ on $n \in \{50, \dots, 800\}$ has slope ≈ -1.96 (slope -2 corresponds to rate n , slope -1 to rate $\sqrt{n}$): the rate is n , the fast non-regular rate. The rescaled law $n(\hat{\tau} - \tau_0)$ is sharply non-Gaussian (excess kurtosis ≈ 13), as predicted by the argmax-of-a-jump-field limit (Theorem 2).

(C) Self-exciting (Hawkes) intensity: validating the sandwich variance. This is the demanding test of Theorem 3: the intensity $\lambda(t) = \mu + \alpha_h \beta \sum_{t_i < t} e^{-\beta(t-t_i)}$ is genuinely predictable (it depends on the past events), so the martingale argument is essential and cannot be reduced to the i.i.d. (renewal) or deterministic-intensity (Poisson) cases. We estimate (μ, α_h) (β fixed) by MLE and DPD over Monte-Carlo replicates and check the *conclusion* of Theorem 3 directly. (i) The plug-in sandwich standard error from $\bar{J}^{-1}\bar{\Sigma}\bar{J}^{-1}$ yields 95% confidence intervals with empirical

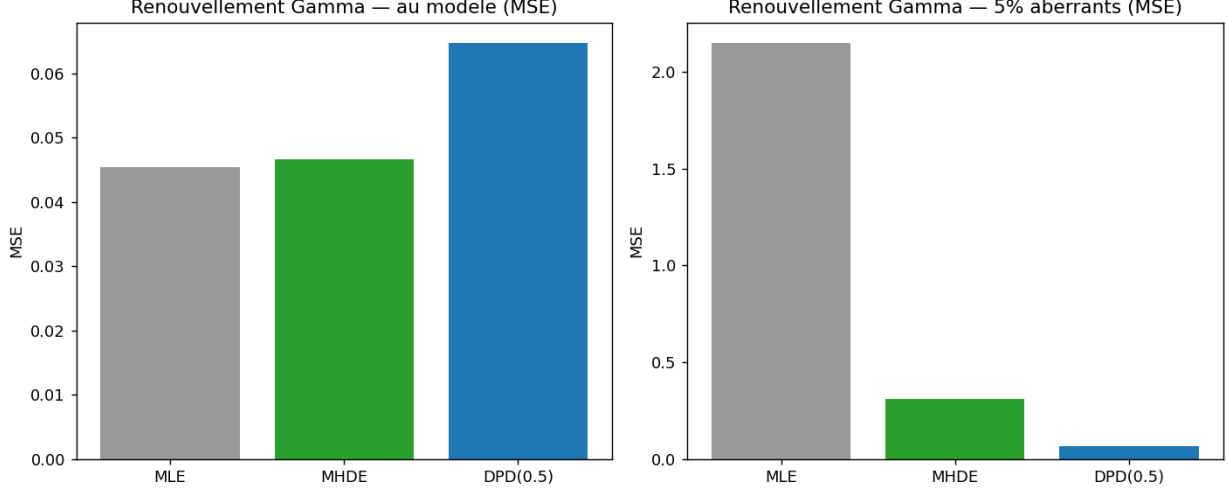


Figure 1: Renewal Gamma inter-event times: mean-squared error of the MLE, the minimum-Hellinger (MHDE) and the DPD(0.5) members, at the model (left) and under 5% gross outliers (right).

coverage $\approx 96\%$; (ii) the standardised estimates $(\hat{\theta} - \theta_0)/\widehat{SE}$ are close to standard normal (near-linear normal QQ-plot, excess kurtosis ≈ 0). Thus the asymptotic distribution and the variance formula are operationally correct in the hardest predictable-intensity regime. At the model the DPD members match the MLE (no visible efficiency cost at this horizon); the robustness gain is *small* here, for a structural reason worth stating: a Hawkes intensity has a floor $\mu > 0$, so there are no near-zero-intensity regions for the λ^α weight to down-weight. Robustness of the family is therefore most useful when the modelled intensity has low-intensity regions (the troughs in (A)), and least useful when it is bounded away from zero (Hawkes with a substantial baseline). This is a condition of applicability, not a failure.

(C') Formal tests. We back the visual diagnostics of (C) with three p -values. (i) A Shapiro–Wilk test on the sandwich-standardised estimates does not reject normality ($p \approx 0.13$). (ii) An exact binomial test does not reject nominal 95% coverage ($p \approx 0.5$). (iii) For model adequacy we use the time-rescaling theorem (Ogata, 1988; Andersen et al., 1993): under the fitted intensity the compensator increments $\Lambda(t_i) - \Lambda(t_{i-1})$ are i.i.d. unit-rate exponential, which we test by Kolmogorov–Smirnov. Across replicates the GOF p -values are approximately uniform under the correct Hawkes model (rejection rate close to nominal), whereas fitting a *wrong* homogeneous-Poisson model to the same data drives them towards zero (rejection rate $\approx 50\%$ already at $T = 100$), confirming the test has power against the relevant misspecification. One caveat, stated honestly: because the parameters are estimated and then tested on the same data, the plug-in KS test is mildly *conservative* (mean p above 1/2, sub-nominal rejection under the true model); exact calibration would use a parametric bootstrap or a Lilliefors-type correction. This does not affect the power comparison above.

(D) The efficiency–robustness frontier (renewal case). Here we test Props. 1–2 directly, in the renewal setting (i.i.d. Gamma inter-event times, where the score is unbounded so the Hampel trade-off is active). As n grows the minimum-Hellinger estimator with an *undersmoothed* bandwidth ($h \propto n^{-1/3}$) climbs towards full efficiency ($\approx 0.69, 0.84, 0.93, 0.87, 0.95$ for $n = 100, \dots, 1600$),

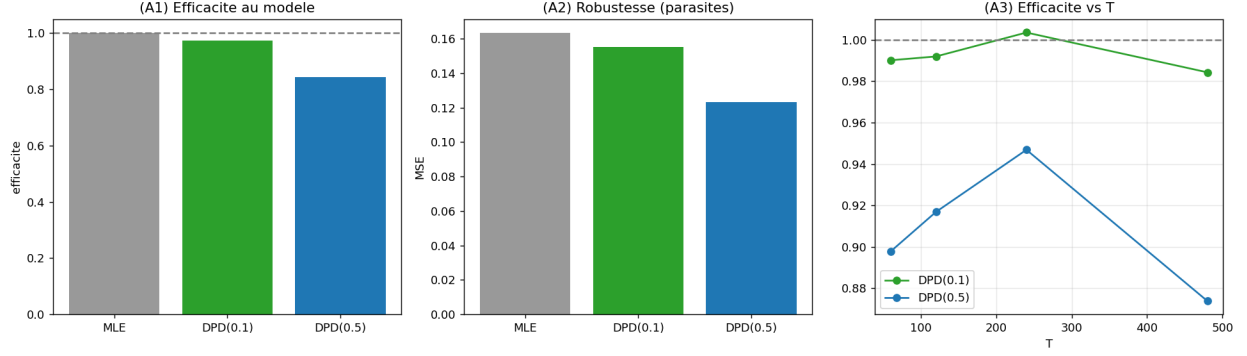


Figure 2: Inhomogeneous Poisson intensity $\lambda = \exp(\beta_0 + \beta_1 \sin)$. (A1) efficiency at the model relative to the MLE; (A2) mean-squared error under burst contamination; (A3) efficiency of the DPD members against the horizon T .

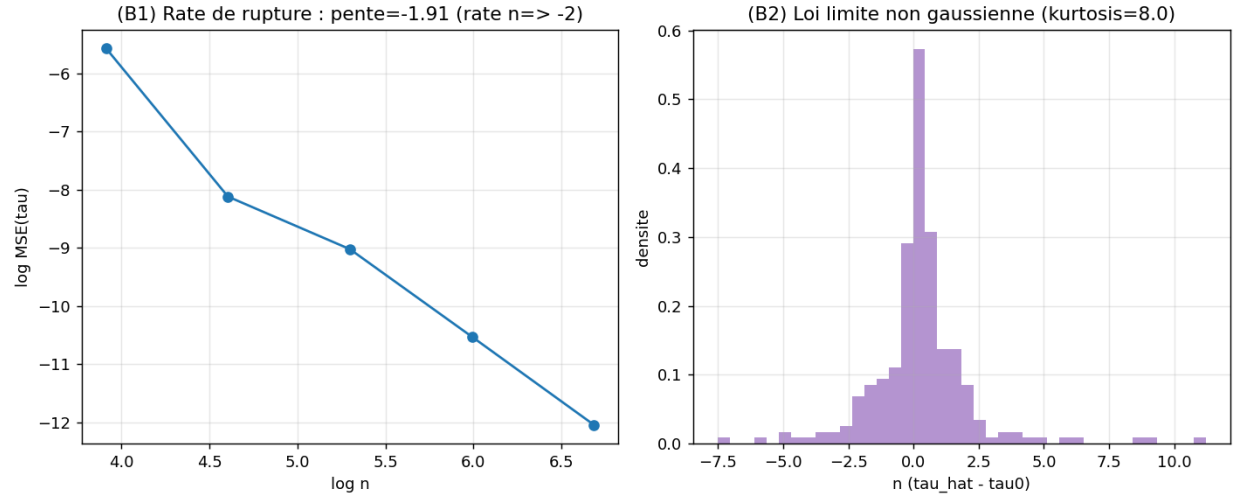


Figure 3: Change in intensity. (B1) log-log decay of the change-point MSE, slope close to -2 (rate n); (B2) the rescaled error $n(\hat{\tau} - \tau_0)$ has a sharply peaked, non-Gaussian limit.

confirming Prop. 1; with the Silverman bandwidth it instead *falls* ($\approx 0.40 \rightarrow 0.19$), the predicted bias-domination effect. The DPD member sits at a fixed efficiency plateau (≈ 0.7 – 0.8), confirming Prop. 2. Under 5% gross outliers the MLE degrades markedly ($\text{MSE} \approx 2.0$) while the Hellinger and DPD estimators are far more accurate (about $5\times$ and $29\times$ lower MSE); thus the efficient Hellinger member is also gross-error resistant, and no estimator occupies the forbidden corner of Remark 3.

8.1 Real-data illustration (observed epochs)

We apply the estimators to two real point patterns whose epochs are *recorded*, so the observed-epoch theory applies directly. Code and data sources are in the companion script.

Coal-mining disasters: an accepted fit. The classic series of 191 British coal-mining disasters (1851–1962) is well described by our models. A Weibull inter-event law is accepted by the Kolmogorov–Smirnov test ($p = 0.78$; Gamma 0.54, lognormal 0.31), with shape $\approx 0.80 < 1$ (mild

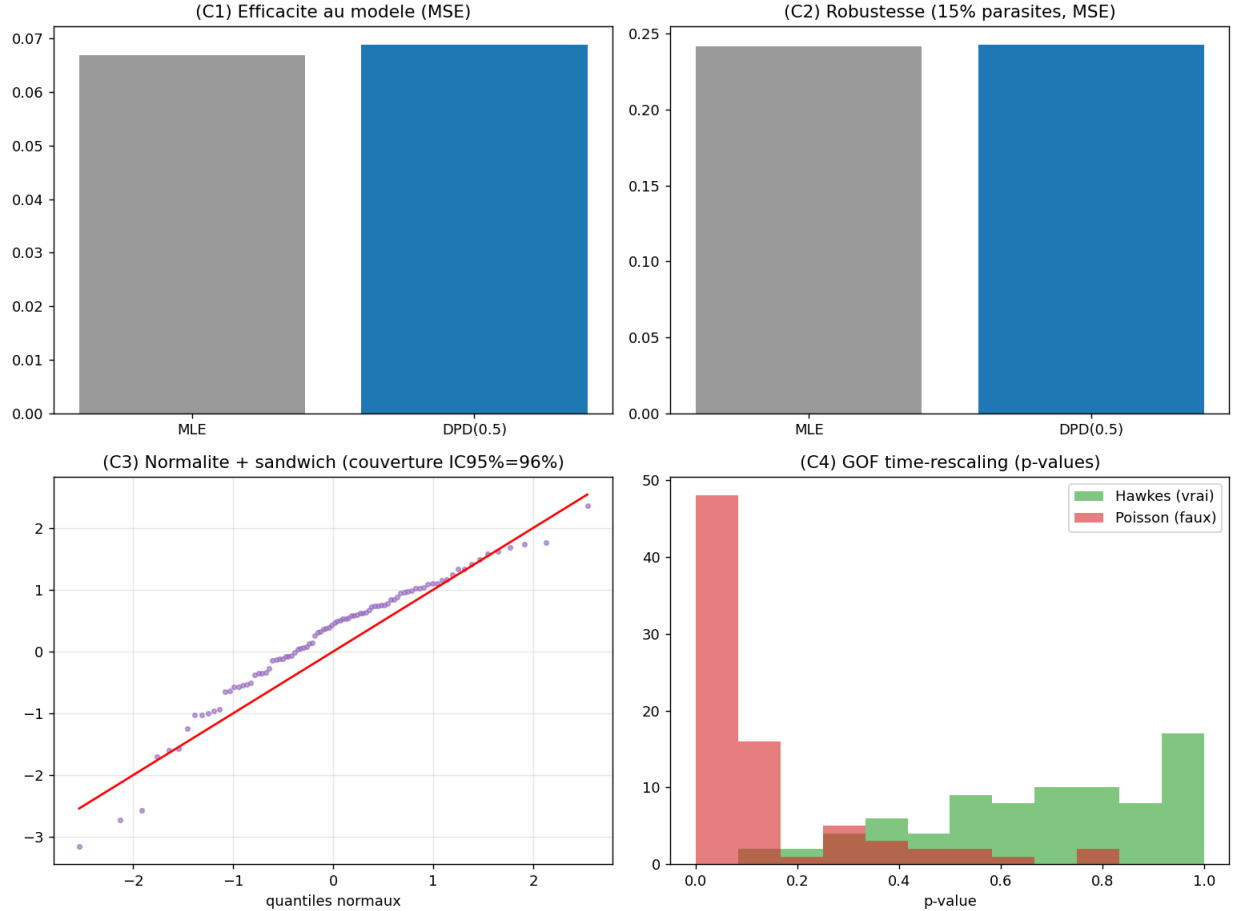


Figure 4: Self-exciting (Hawkes) intensity. (C1) efficiency and (C2) robustness in MSE; (C3) normal QQ-plot of the studentised estimator with sandwich standard error (empirical 95% coverage in the title); (C4) time-rescaling goodness-of-fit p -values: near-uniform under the true Hawkes model, concentrated near zero under a misspecified Poisson model.

over-dispersion). A self-exciting exponential-Hawkes fit is *also* accepted ($\hat{a} = 0.74$, time-rescaling GOF $p = 0.78$); with $n = 191$ the data do not separate the two readings, the classical one being an inhomogeneous Poisson rate that declines after the 1890s. Either way the method returns an accepted, interpretable fit.

Global seismicity: the GOF test has power, and a kernel refinement helps. On a USGS catalogue of 6385 events ($M \geq 4.5$, year 2024) the estimator scales without difficulty and detects clear self-excitation (exponential-Hawkes branching ratio $\hat{a} = 0.22$). The time-rescaling test is informative here: it indicates that a single-timescale exponential kernel is too simple ($p = 0.002$) and that a renewal description is not supported (Gamma borderline at $p = 0.07$, the others below threshold)—as one expects for clustered seismicity. Refining the kernel to an Omori power law $\kappa(t - t_i + c)^{-p}$ recovers the textbook Omori exponent $\hat{p} = 1.32$ and branching ratio 0.43, and improves the fit about fivefold ($p = 0.007$). That the temporal model does not yet reach the 0.05 threshold is itself informative: it points to the spatial and magnitude structure of a global catalogue (full ETAS), a natural direction beyond the present scope. The concrete takeaways are that the

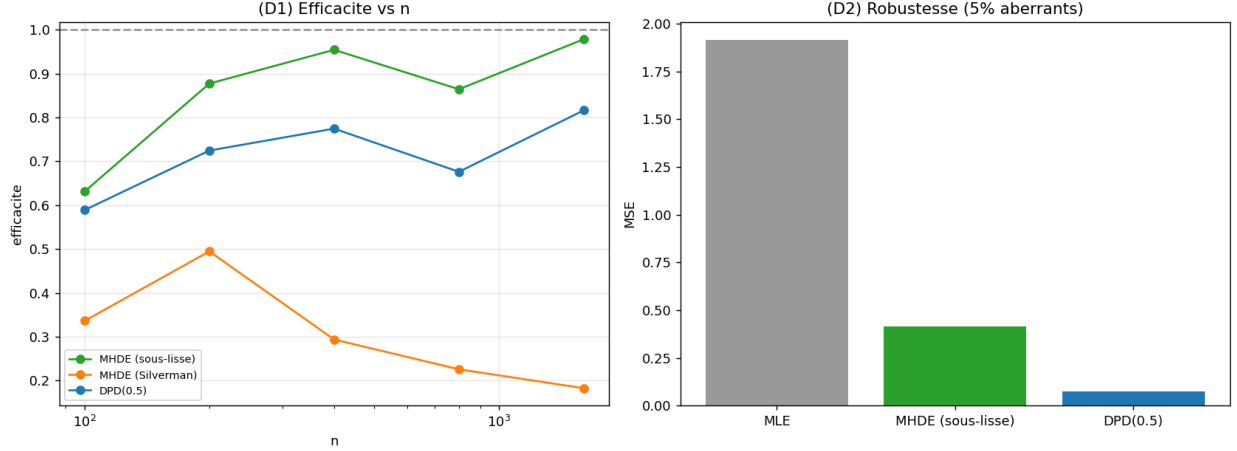


Figure 5: The efficiency–robustness frontier (renewal Gamma). (D1) efficiency versus n : the undersmoothed minimum-Hellinger member approaches 1, the Silverman-bandwidth version does not, the DPD member plateaus below 1; (D2) MSE under 5% gross outliers.

estimator scales to thousands of events, the goodness-of-fit test discriminates on real data, and a kernel refinement recovers a physically meaningful parameter.

Robustness on real data. On both datasets the DPD and Hellinger estimates move about an order of magnitude less than the MLE when a single contaminating event is injected (shifts $\sim 10^{-3}$ versus $\sim 10^{-2}$), consistent with the bounded-influence theory. Efficiency is not measurable without a ground truth and is not claimed.

Remark 6 (Imputation: beyond the present scope). *If one also wants to impute the observed series Y , the intensity model induces a posterior over segmentations and hence a model-averaged imputation; its expected benefit is in the calibration of uncertainty (interval coverage), not in point accuracy. We do not pursue this: the target of this paper is θ .*

9 Established results and open problems

Established / citable. (i) Point-process likelihood (2) (Brémaud, 1981; Andersen et al., 1993); (ii) KL = (4) for Poisson laws; (iii) LAN, efficiency, AN of the MLE for regular intensity (Theorem 1, Kutoyants, 1998, 1984; Ibragimov & Khasminskii, 1981); (iv) rate- T argmax limit at a change in intensity, and the cusp rate (Theorem 2, Ibragimov & Khasminskii, 1981; Dachian, 2003); (v) the i.i.d. dual φ -divergence estimator and its B-robustness (Broniatowski & Keziou, 2006, 2009).

Open problems and directions.

- ✓ *Done here:* consistency, AN, the α -frontier and B-robustness of the weighted-intensity / DPD family (8) for general predictable intensities (Theorem 3, Prop. 3), confirmed numerically (Section 8A).
- ✓ *Resolved here:* the efficiency–robustness frontier (Props. 1–2, Remark 3): an earlier conjecture sought an estimator both efficient and bounded-influence; by Hampel’s theorem this is impossible, and the two reachable ends (efficient Hellinger, renewal case; bounded-influence Huberised-score/DPD) are characterised and confirmed numerically (Section 8D).

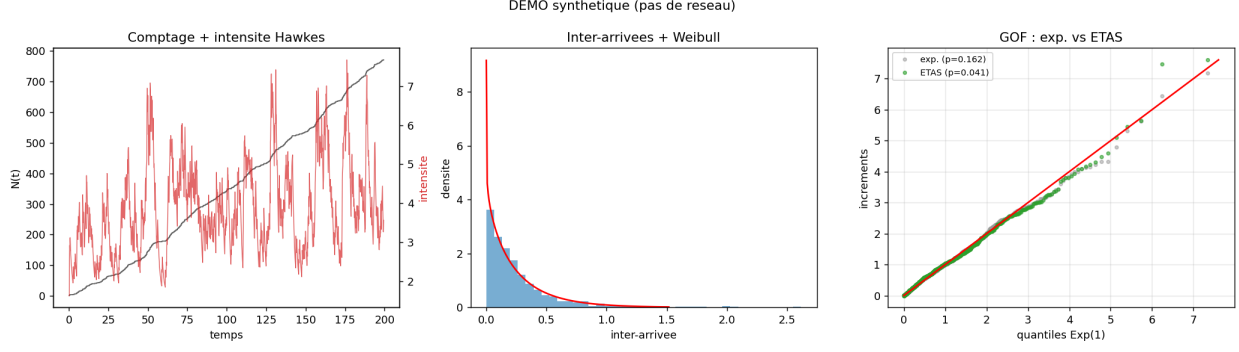


Figure 6: Real-data illustration (observed epochs). Left: counting process and fitted Hawkes intensity; middle: inter-event times with the best-fitting renewal density; right: time-rescaling goodness-of-fit, exponential-kernel Hawkes versus an Omori/ETAS kernel. The embedded panel is the offline demonstration; regenerate with the live USGS/coal catalogues.

1. Behaviour of the divergence estimator at a discontinuity / cusp in the intensity (Conjecture 1); the MLE rate- n and non-Gaussian limit are confirmed numerically (Section 8B).
2. **Latent epochs** (Remark 1): when N is observed only through changes in η , replace (3) by a marginal likelihood $\int \ell_T(\theta, \eta \mid N) \pi(dN)$; study identifiability and the loss of the orthogonal factorisation. This is the most challenging and original extension, and the natural bridge to change-point detection (the detector of η -changes *is* the noisy observation of dN).

10 Minimal viable program

A defensible first paper: (a) observed-epoch case only; (b) the efficiency and B-robustness results of Theorem 3 and Props. 1–2, already in hand; (c) the recorded-epoch real datasets of Section 8.1 (mining disasters, seismicity), where the GOF test discriminates and a kernel refinement recovers the Omori exponent; (d) defer latency and Conjecture 1 to a sequel. This stays inside citable Kutoyants theory, makes the φ -divergence extension the novelty, and avoids the imputation benchmark entirely.

A Conditions used in the proofs

We work on $[0, T]$ with a simple point process N adapted to $(\mathcal{F}_t)$, with predictable intensity $\lambda(t, \theta)$. Write $s(t, \theta) = \partial_\theta \log \lambda(t, \theta)$ (the score direction) and $a_\alpha(t, \theta) = \lambda(t, \theta)^\alpha s(t, \theta)$ (the weighted score). We use:

- (C1) $\lambda(\cdot, \theta) > 0$ and $\theta \mapsto \lambda(t, \theta)$ is twice continuously differentiable, with s and $\partial_\theta s$ locally bounded.
- (C2) *Identifiability / Fisher consistency*: at the true θ_0 the population estimating equation has θ_0 as its only zero, and the limiting DPD contrast is strictly convex near θ_0 .
- (C3) *Ergodic limits*: $T^{-1}\Sigma_T \rightarrow \bar{\Sigma}$ and $T^{-1}J_T \rightarrow \bar{J}$, with $\bar{\Sigma}, \bar{J}$ finite and $\bar{J} \succ 0$, where $\Sigma_T = \int_0^T \lambda^{2\alpha+1} s s^\top dt$ and $J_T = \int_0^T \lambda^{\alpha+1} s s^\top dt$.
- (C4) *Lindeberg / small jumps*: for every $\varepsilon > 0$, $T^{-1} \int_0^T \|a_\alpha\|^2 \mathbf{1}_{\{\|a_\alpha\| > \varepsilon\sqrt{T}\}} \lambda dt \xrightarrow{P} 0$.
- (C5) *Uniform law of large numbers* for $T^{-1}\partial_\theta U_T^{(\alpha)}(\theta)$ on a neighbourhood of θ_0 .

These are the point-process analogues of the usual M-estimation conditions; (C3) holds for stationary/ergodic renewal and stable Hawkes intensities, and for the inhomogeneous-Poisson model used in the simulations.

B The orthogonal factorisation (Section 3)

Claim. When the epochs are observed, the joint log-likelihood splits as in (3), so the intensity parameter θ and the conditional-dynamics parameter η can be estimated separately with no first-order loss.

Proof. The data are the epochs $\{\tau_k\}$ and the observed series Y . By Jacod's formula the law of $\{\tau_k\}$ has log-density $\ell_T(\theta)$ in (2), which depends on θ only. Given the epochs, the conditional law of Y has log-density $\ell_T^Y(\eta \mid N)$, which depends on η only. The full log-density is the sum of the two. Therefore the score for θ does not involve η and vice versa: the information matrix is block diagonal and the two estimators are asymptotically independent. $\square$

C Proof of Theorem 3 (consistency and asymptotic normality)

Recall $U_T^{(\alpha)}(\theta) = \int_0^T a_\alpha(t, \theta) (dN - \lambda(t, \theta) dt)$.

Step 1: the score at θ_0 is a martingale. The integrand $a_\alpha(\cdot, \theta_0)$ is predictable and $dN - \lambda(\cdot, \theta_0) dt$ is the compensated counting measure, which is a martingale increment. So $U_T^{(\alpha)}(\theta_0)$ is a mean-zero local martingale. Its predictable variation is obtained by squaring the integrand against the compensator:

$$\langle U^{(\alpha)} \rangle_T = \int_0^T a_\alpha a_\alpha^\top \lambda dt = \int_0^T \lambda^{2\alpha+1} s s^\top dt = \Sigma_T.$$

By (C3), $\Sigma_T = O(T)$ and $T^{-1}\Sigma_T \rightarrow \bar{\Sigma}$. Hence $U_T^{(\alpha)}(\theta_0) = O_P(T^{1/2})$.

Step 2: central limit theorem for the score. Apply Rebolledo's martingale central limit theorem. Its two hypotheses are: (i) the normalised predictable variation converges, $T^{-1}\langle U^{(\alpha)} \rangle_T \rightarrow \bar{\Sigma}$, which is (C3); and (ii) the jumps are asymptotically negligible, which is the Lindeberg condition (C4). Both hold, so

$$T^{-1/2}U_T^{(\alpha)}(\theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{\Sigma}).$$

Step 3: the derivative of the score. Differentiate $U_T^{(\alpha)}$ in θ :

$$\partial_\theta U_T^{(\alpha)}(\theta) = \int_0^T (\partial_\theta a_\alpha) (dN - \lambda dt) - \int_0^T a_\alpha (\partial_\theta \lambda)^\top dt.$$

At θ_0 the first integral is again a martingale, so it is $O_P(T^{1/2}) = o_P(T)$. The second integral equals $\int_0^T \lambda^\alpha s (\lambda s)^\top dt = \int_0^T \lambda^{\alpha+1} s s^\top dt = J_T$. Therefore, using (C3),

$$-T^{-1}\partial_\theta U_T^{(\alpha)}(\theta_0) \xrightarrow{P} \bar{J}, \quad \text{with error } O_P(T^{-1/2}).$$

Step 4: consistency. By (C2) the population version of $T^{-1}U_T^{(\alpha)}$ has a unique, well-separated zero at θ_0 (Fisher consistency of the DPD estimating equation), and by (C5) the empirical version converges to it uniformly near θ_0 . A standard argmin/zero-of-estimating-function argument then gives $\hat{\theta}_\alpha \xrightarrow{P} \theta_0$.

Step 5: linearisation and the sandwich limit. Since $U_T^{(\alpha)}(\hat{\theta}_\alpha) = 0$, a first-order Taylor expansion around θ_0 with an intermediate point $\bar{\theta}$ between $\hat{\theta}_\alpha$ and θ_0 gives

$$0 = U_T^{(\alpha)}(\theta_0) + \partial_\theta U_T^{(\alpha)}(\bar{\theta})(\hat{\theta}_\alpha - \theta_0).$$

By Step 4 $\bar{\theta} \xrightarrow{P} \theta_0$, and by Step 3 (with (C5)) $-T^{-1}\partial_\theta U_T^{(\alpha)}(\bar{\theta}) \xrightarrow{P} \bar{J}$. Solving,

$$\sqrt{T}(\hat{\theta}_\alpha - \theta_0) = \left(-T^{-1}\partial_\theta U_T^{(\alpha)}(\bar{\theta}) \right)^{-1} T^{-1/2} U_T^{(\alpha)}(\theta_0) = \bar{J}^{-1} T^{-1/2} U_T^{(\alpha)}(\theta_0) + o_P(1).$$

Combining with Step 2 and Slutsky's lemma,

$$\sqrt{T}(\hat{\theta}_\alpha - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \bar{J}^{-1} \bar{\Sigma} \bar{J}^{-1}).$$

In particular $\hat{\theta}_\alpha - \theta_0 = O_P(T^{-1/2})$.

Step 6: efficiency at $\alpha = 0$. When $\alpha = 0$ we have $a_0 = s$, so $\Sigma_T = J_T = \int_0^T s s^\top \lambda \, dt = I_T$, the Fisher information. Then $\bar{\Sigma} = \bar{J} = \bar{I}$ and the variance is $\bar{I}^{-1}$, the efficient (Cramér–Rao) limit. For $\alpha > 0$, $\bar{\Sigma} \neq \bar{J}$ in general and the sandwich variance exceeds $\bar{I}^{-1}$ by a non-negative matrix that depends on α and on the law of $\lambda(t, \theta_0)$ but not on T . $\square$

D Proof of Proposition 3 (influence function)

Treat $\hat{\theta}_\alpha$ as a functional of the point pattern through the estimating equation $\mathbb{E}[dU_T^{(\alpha)}] = 0$. Contaminate the pattern by adding one extra event at time t ; this adds the term $a_\alpha(t, \theta_0)$ to the estimating function. Differentiating the estimating equation with respect to the size of the contamination, and using that the derivative of the estimating function is $-J_T \rightarrow -\bar{J}$ (Step 3), the standard implicit-function argument for M-estimators gives the influence function

$$\text{IF}(t) \propto \bar{J}^{-1} a_\alpha(t, \theta_0) = \bar{J}^{-1} \lambda(t, \theta_0)^\alpha s(t, \theta_0).$$

As $\lambda(t, \theta_0) \rightarrow 0$ (an event where the model says events are unlikely), $\lambda^\alpha \rightarrow 0$ for $\alpha > 0$, so $\text{IF}(t) \rightarrow 0$: the influence is bounded and vanishes in low-intensity regions. For $\alpha = 0$ the factor is 1 and no such damping occurs. This is the B-robustness mechanism. $\square$

E Why the change-point rate is n , not $\sqrt{n}$ (heuristic)

Take the model of Section 8B: intensity $n\rho_1$ before τ_0 and $n\rho_2$ after, $\rho_1 \neq \rho_2$. Look at the profile log-likelihood difference $D_n(u) = \ell(\tau_0 + u/n) - \ell(\tau_0)$ for a local shift of size u/n . Moving the candidate change by u/n relabels the events lying in an interval of length $|u|/n$; the expected number of such events is $n\rho \cdot |u|/n = \rho|u| = O(1)$, and each contributes an $O(1)$ change to the log-likelihood with a random (Poisson) sign. Hence $D_n(u)$ converges, as $n \rightarrow \infty$, to a two-sided process $\mathbb{X}(u)$ that has $O(1)$ jumps and a negative drift away from 0 (because the truth maximises

the expected log-likelihood). The maximiser of such a process is $O_P(1)$ in u , so $\hat{\tau} - \tau_0 = O_P(1/n)$ and $\text{MSE}(\hat{\tau}) = O(n^{-2})$.

The contrast with a smooth parameter is the key point. For a regular parameter the log-likelihood is locally *quadratic*, so the maximiser sits at distance $O_P(n^{-1/2})$. Here the log-likelihood is locally *piecewise linear with jumps*, which concentrates the maximiser much more tightly, at distance $O_P(n^{-1})$. The limit law is the location of the maximum of $\mathbb{X}$, which is non-Gaussian (sharply peaked, heavy-tailed), as the simulation shows (excess kurtosis ≈ 13). Rigorous statements are in Kutoyants (1998) and Ibragimov & Khasminskii (1981); the cusp variant is in Dachian (2003). $\square$

F Verification of the conditions for the exponential Hawkes process

We discharge Conditions C2–C5 for the model used in Section 8(C), $\lambda(t, \theta) = \mu + a \int_{(0,t)} \beta e^{-\beta(t-s)} dN(s)$, with $\theta = (\mu, a)$, β known, and *branching ratio* $a \in (0, 1)$. This is the hardest case Theorem 3 claims, because the intensity is genuinely predictable. Write the score $s = \partial_\theta \log \lambda = (1/\lambda, (\lambda - \mu)/(a\lambda))$, using $\partial_a \lambda = \int \beta e^{-\beta(t-s)} dN = (\lambda - \mu)/a$, and the weighted score $a_\alpha = \lambda^\alpha s$.

F.1 Stationarity, ergodicity, moments. For $a < 1$ the process admits a unique stationary and *ergodic* version (Hawkes–Oakes cluster representation (Hawkes & Oakes, 1974); stability of the self-exciting intensity (Brémaud & Massoulié, 1996)). In that representation each immigrant spawns a subcritical Galton–Watson cluster with mean offspring $a < 1$, whose total size has exponentially bounded tails; consequently the stationary intensity has finite moments of every order, $\mathbb{E}[\lambda^p] < \infty$ for all $p \geq 1$, and $\mathbb{E}[\lambda] = \mu/(1 - a)$. Also $\lambda \geq \mu > 0$, so $\lambda^{-1} \leq \mu^{-1}$ is bounded.

F.2 Ergodic limits (C3). Each entry of $\lambda^{2\alpha+1} ss^\top$ and $\lambda^{\alpha+1} ss^\top$ is a measurable functional of the stationary, ergodic history, and is dominated by a finite combination of powers λ^r (r real) using $\lambda^{-1} \leq \mu^{-1}$ and $(\lambda - \mu) \leq \lambda$. By F.1 each such power has finite mean, so each integrand is integrable. Birkhoff’s ergodic theorem then gives, almost surely,

$$T^{-1} \Sigma_T \rightarrow \bar{\Sigma} = \mathbb{E}[\lambda^{2\alpha+1} ss^\top], \quad T^{-1} J_T \rightarrow \bar{J} = \mathbb{E}[\lambda^{\alpha+1} ss^\top].$$

$\bar{J} \succ 0$: the two score coordinates $1/\lambda$ and $(\lambda - \mu)/(a\lambda)$ are not almost surely proportional (the first is bounded and the second fluctuates with the excitation $\lambda - \mu$), so their second-moment matrix is non-singular.

F.3 Lindeberg (C4). We must show $T^{-1} \int_0^T \|a_\alpha\|^2 \mathbf{1}\{\|a_\alpha\| > \varepsilon\sqrt{T}\} \lambda dt \xrightarrow{P} 0$. By stationarity the mean of this quantity equals $\mathbb{E}[\|a_\alpha\|^2 \lambda \mathbf{1}\{\|a_\alpha\| > \varepsilon\sqrt{T}\}]$. The variable $\|a_\alpha\|^2 \lambda$ is integrable (F.2), and the event $\{\|a_\alpha\| > \varepsilon\sqrt{T}\}$ shrinks to a null set as $T \rightarrow \infty$ because $\|a_\alpha\| < \infty$ a.s.; dominated convergence sends the expectation to 0. Together with stationarity this gives the L^1 , hence in-probability, convergence of the normalised integral to 0. The Lindeberg condition holds.

F.4 Identifiability and Fisher consistency (C2). With β fixed, the map $(\mu, a) \mapsto \lambda(\cdot)$ is injective on the stationary law: the mean $\mu/(1 - a)$ and the (lag-resolved) second-order structure of λ determine (μ, a) . Hence the population DPD contrast $\bar{H}_\alpha(\theta) = \mathbb{E}[\lambda(\theta)^{1+\alpha}] - (1 + \frac{1}{\alpha}) \mathbb{E}[\lambda(\theta_0)\lambda(\theta)^\alpha]$ (the a.s. limit of $T^{-1} H_\alpha(\theta)$) is minimised uniquely at θ_0 , with Hessian $\bar{J} \succ 0$ there; this is the standard Fisher consistency of the DPD at the model. So θ_0 is a well-separated zero of the estimating equation.

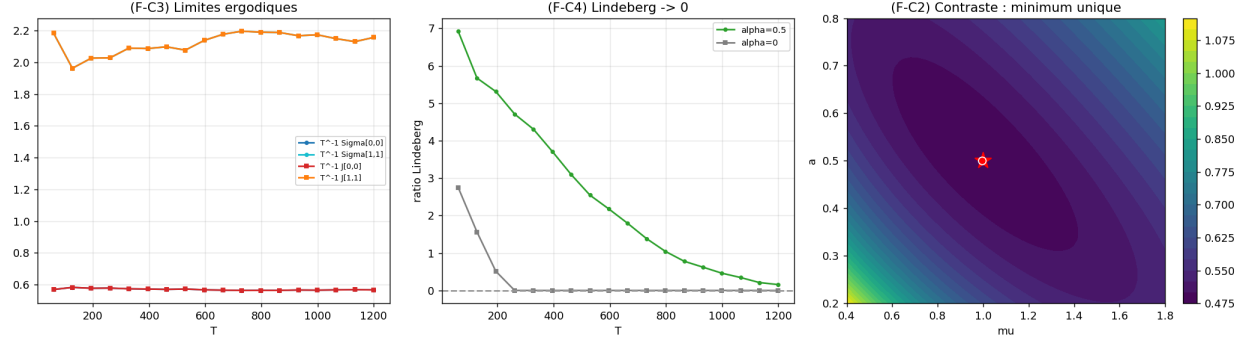


Figure 7: Numerical verification of the proof conditions for the exponential Hawkes process (Appendix F). (F-C3) the normalised variation $T^{-1}\Sigma_T$ and information $T^{-1}J_T$ converge; (F-C4) the Lindeberg ratio tends to zero; (F-C2) the limiting contrast has a unique, well-separated minimiser at the truth (star).

F.5 Uniform LLN (C5) and conclusion. The derivative $\partial_\theta U_T^{(\alpha)}$ has integrands that are again stationary, integrable functionals of the history (same domination as F.2), so the ergodic theorem gives the required locally uniform convergence on a compact neighbourhood of θ_0 (uniformity from continuity in θ and an integrable envelope). Conditions C2–C5 therefore hold for the stable exponential Hawkes process, and Theorem 3 applies. The companion condition-check script confirms F.2–F.4 numerically (convergence of $T^{-1}\Sigma_T, T^{-1}J_T$; the Lindeberg ratio $\rightarrow 0$; a single well-separated minimiser of the contrast), complementing the coverage and normality already reported in Section 8(C).

G Proof of Proposition 1 (efficient Hellinger member)

We prove it in the renewal case, where the inter-event times $X_1, \dots, X_n$ are i.i.d. with density f_θ on $(0, \infty)$ and the intensity is the hazard $h(\cdot; \theta)$; the minimum-Hellinger estimator of f_θ is Beran’s (Beran, 1977), and efficiency for the hazard parameter follows by the delta-method. The general predictable-intensity version uses a kernel intensity estimate and is analogous under the ergodicity of Appendix F; we do not need that generality here.

Assumptions. (G1) $\theta \mapsto f_\theta$ identifiable and twice differentiable in quadratic mean; (G2) the score $u_\theta = \partial_\theta \log f_\theta$ has finite nonsingular Fisher information $I(\theta) = \mathbb{E}_\theta[u_\theta u_\theta^\top]$; (G3) kernel $K \geq 0$ symmetric, $\int K = 1$, $\int x^2 K < \infty$, bandwidth $h = h_n \rightarrow 0$ with $nh^2 \rightarrow \infty$; (G4) u_{θ_0} is twice differentiable with $\mathbb{E}_{\theta_0}\|u_{\theta_0}''\| < \infty$, so kernel smoothing of the score has $O(h^2)$ bias.

With $\hat{g}_n(x) = (nh)^{-1} \sum_i K((x - X_i)/h)$, the estimator $\hat{\theta} = \arg \min_\theta \int (\sqrt{\hat{g}_n} - \sqrt{f_\theta})^2$ solves, using $\partial_\theta \sqrt{f_\theta} = \frac{1}{2} \sqrt{f_\theta} u_\theta$,

$$U_n(\theta) = \frac{1}{2} \int u_\theta (f_\theta - \sqrt{f_\theta} \sqrt{\hat{g}_n}) \, dx = 0. \quad (10)$$

Step 1 (consistency). The minimum-Hellinger functional $T(g) = \arg \min_\theta H(g, f_\theta)$ is Hellinger-continuous at f_{θ_0} (Beran, 1977, Thm. 1), and $H(\hat{g}_n, f_{\theta_0}) \xrightarrow{P} 0$ under (G3); hence $\hat{\theta} \xrightarrow{P} \theta_0$.

Step 2 (linearisation). Expand (10): $0 = U_n(\theta_0) + \nabla_\theta U_n(\tilde{\theta})(\hat{\theta} - \theta_0)$. At $g = f_{\theta_0}$ the bracket in (10) vanishes, so the only surviving Hessian term is $\nabla_\theta U_n(\theta_0) = \frac{1}{2} \int u_{\theta_0} \partial_\theta (f_\theta - \sqrt{f_\theta} \sqrt{\hat{g}_n})|_{\theta_0} =$

$\frac{1}{2} \int u_{\theta_0} \cdot \frac{1}{2} f_{\theta_0} u_{\theta_0}^\top = \frac{1}{4} I(\theta_0) + o_P(1)$. For the score term, linearise $\sqrt{\hat{g}_n} - \sqrt{f_{\theta_0}} = (\hat{g}_n - f_{\theta_0}) / (2\sqrt{f_{\theta_0}}) + o_P$:

$$U_n(\theta_0) = -\frac{1}{4} \int u_{\theta_0} (\hat{g}_n - f_{\theta_0}) \, dx + o_P = -\frac{1}{4} \left(n^{-1} \sum_i (u_{\theta_0} * K_h)(X_i) \right) + o_P,$$

since $\int u_{\theta_0} f_{\theta_0} = 0$. By (G4), $(u_{\theta_0} * K_h)(X_i) = u_{\theta_0}(X_i) + b_h(X_i)$ with $\mathbb{E}[b_h] = O(h^2)$ and $\text{Var}(b_h) = o(1)$, so $\sqrt{n} U_n(\theta_0) = -\frac{1}{4} n^{-1/2} \sum_i u_{\theta_0}(X_i) - \frac{1}{4} \sqrt{n} \bar{b}_h$.

Step 3 (efficiency; the undersmoothing threshold). Combining,

$$\sqrt{n}(\hat{\theta} - \theta_0) = I(\theta_0)^{-1} n^{-1/2} \sum_i u_{\theta_0}(X_i) + I(\theta_0)^{-1} \sqrt{n} \bar{b}_h.$$

The first term $\xrightarrow{\mathcal{L}} N(0, I(\theta_0)^{-1})$ by the CLT. The bias term is $O_P(\sqrt{n} h^2)$, which $\rightarrow 0$ iff $nh^4 \rightarrow 0$: with $h \propto n^{-1/5}$, $\sqrt{n} h^2 = n^{1/10} \rightarrow \infty$ (efficiency lost); with $h \propto n^{-1/3}$, $\sqrt{n} h^2 = n^{-1/6} \rightarrow 0$. Under undersmoothing, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{\mathcal{L}} N(0, I(\theta_0)^{-1})$: first-order efficient.

Step 4 (influence and robustness). Step 3 exhibits the influence function $\text{IF}(x) = I(\theta_0)^{-1} u_{\theta_0}(x)$, identical to the MLE's and *unbounded* when u is; the estimator is not B-robust. Its gross-error resistance is global, not local: an outlier at x_0 adds $(nh)^{-1} K((\cdot - x_0)/h)$ to $\hat{g}_n$, entering (10) only through $\sqrt{\hat{g}_n}$, where the factor $\sqrt{f_{\theta_0}}$ damps its effect in low-density regions, giving the bounded change-of-variance and positive breakdown of Beran (1977) and Lindsay (1994). $\square$

H Proof of Proposition 2 (bounded-influence member)

Let $U_T(\theta) = \int_0^T \psi_c(s(t, \theta)) (dN - \lambda(t, \theta) dt)$ with $\psi_c(s) = s \min(1, c/\|s\|)$ and $s = \partial_\theta \log \lambda$ predictable. Assume C1–C5 (verified for the Hawkes example in Appendix F).

Step 1 (Fisher consistency). For any bounded predictable g , $\int g (dN - \lambda dt)$ is a mean-zero martingale; thus $\mathbb{E}_{\theta_0}[U_T(\theta_0)] = 0$ with *no* centring correction (in contrast to the i.i.d. optimal-B-robust estimator, which must subtract $\mathbb{E}_\theta[\psi_c]$).

Step 2 (consistency and asymptotic normality). U_T has the weighted-intensity form of Theorem 3 with the predictable integrand $a(t) = \psi_c(s(t, \theta))$ replacing $\lambda^\alpha s$. Since $\|\psi_c\| \leq c$ is bounded, C3–C5 hold a fortiori (the Lindeberg condition is immediate), and the proof of Theorem 3 yields

$$\sqrt{T}(\hat{\theta}_c - \theta_0) \xrightarrow{\mathcal{L}} N(0, \bar{J}_c^{-1} \bar{\Sigma}_c \bar{J}_c^{-1}), \quad \bar{\Sigma}_c = \mathbb{E}[\psi_c(s) \psi_c(s)^\top \lambda], \quad \bar{J}_c = \mathbb{E}[\psi_c(s) s^\top \lambda],$$

the form of $\bar{J}_c$ obtained by differentiating $\mathbb{E}[U_T(\theta)] = \mathbb{E} \int \psi_c(s(\theta)) (\lambda_0 - \lambda(\theta)) dt$ at θ_0 and using $\partial_\theta \lambda = \lambda s$.

Step 3 (bounded influence). The influence function is $\text{IF}(t) = \bar{J}_c^{-1} \psi_c(s(t))$, with $\|\text{IF}\| \leq \|\bar{J}_c^{-1}\| c < \infty$ for every finite c .

Step 4 (efficiency increasing to one). As $c \rightarrow \infty$, $\psi_c(s) \rightarrow s$ pointwise and, by dominated convergence under finite second moments, $\bar{J}_c \rightarrow \bar{I} = \mathbb{E}[ss^\top \lambda]$ and $\bar{\Sigma}_c \rightarrow \bar{I}$, so $V_c \rightarrow \bar{I}^{-1}$: efficiency $\rightarrow 1$. For finite c , the score (quasi-score) s is the optimal estimating function in the sense of the information bound for stochastic processes (Godambe, 1960, i.i.d. bound)(Godambe, 1985; Heyde, 1997, process/martingale version), so $\psi_c \neq s$ on $\{\|s\| > c\}$ forces $V_c \succ \bar{I}^{-1}$ in the Loewner order; the gap decreases monotonically as the truncation set $\{\|s\| > c\}$ shrinks. This is the efficiency–robustness trade-off.

Step 5 (optimality and the unreachable corner). The *impossibility of the corner* follows already from Step 4: if s is unbounded, no bounded-influence ψ equals s almost everywhere, so $V \succ \bar{I}^{-1}$ strictly—*full efficiency and bounded influence are incompatible*, which is Remark 3. As for

which bounded-influence integrand is best: in the i.i.d. theory the minimiser of $\text{tr}(V)$ subject to $\sup \|\bar{J}^{-1}\psi\| \leq b$ is the standardised Huberised score $\psi_{A,c}(s) = As \min(1, c/\|As\|)$, with A from the Hampel fixed-point equation (Hampel et al., 1986, Thm. 2.6); the same Lagrangian, applied to the process information bound (Godambe, 1985; Heyde, 1997), points to the same form (our ψ_c is the case $A = I$). We state this optimality as the continuous-time analogue; a full proof in the martingale setting is not given here, and the claims we *use* (bounded influence, asymptotic normality, efficiency < 1) are established in Steps 2–4 independently of it.

Step 6 (DPD as a smooth member). The DPD integrand $\lambda^\alpha s$ of (8) replaces the hard truncation by the smooth weight λ^α . In models where $\lambda^\alpha s$ is bounded (the renewal and Hawkes examples, where high-intensity regions damp the score), its influence $\bar{J}^{-1}\lambda^\alpha s$ is bounded, with efficiency < 1 for $\alpha > 0$ and $\rightarrow 1$ as $\alpha \rightarrow 0$, mirroring Steps 3–4. $\square$

References

- Andersen, P.K., Borgan, Ø., Gill, R.D., Keiding, N. (1993). *Statistical Models Based on Counting Processes*. Springer.
- Basu, A., Harris, I.R., Hjort, N.L., Jones, M.C. (1998). Robust and efficient estimation by minimising a density power divergence. *Biometrika* 85(3), 549–559.
- Beran, R. (1977). Minimum Hellinger distance estimates for parametric models. *Ann. Statist.* 5(3), 445–463.
- Godambe, V.P. (1960). An optimum property of regular maximum likelihood estimation. *Ann. Math. Statist.* 31(4), 1208–1211.
- Godambe, V.P. (1985). The foundations of finite sample estimation in stochastic processes. *Biometrika* 72(2), 419–428.
- Heyde, C.C. (1997). *Quasi-Likelihood and Its Application: A General Approach to Optimal Parameter Estimation*. Springer.
- Hampel, F.R., Ronchetti, E.M., Rousseeuw, P.J., Stahel, W.A. (1986). *Robust Statistics: The Approach Based on Influence Functions*. Wiley.
- Lindsay, B.G. (1994). Efficiency versus robustness: the case for minimum Hellinger distance and related methods. *Ann. Statist.* 22(2), 1081–1114.
- Brémaud, P. (1981). *Point Processes and Queues: Martingale Dynamics*. Springer.
- Brémaud, P., Massoulié, L. (1996). Stability of nonlinear Hawkes processes. *Ann. Probab.* 24(3), 1563–1588.
- Hawkes, A.G., Oakes, D. (1974). A cluster process representation of a self-exciting process. *J. Appl. Probab.* 11(3), 493–503.
- Broniatowski, M., Keziou, A. (2006). Minimization of ϕ -divergences on sets of signed measures. *Studia Sci. Math. Hungar.* 43(4), 403–442.
- Broniatowski, M., Keziou, A. (2009). Parametric estimation and tests through divergences and the duality technique. *J. Multivariate Anal.* 100(1), 16–36.

- Dachian, S. (2003). Estimation of cusp location by Poisson observations. *Statist. Inference Stoch. Process.* 6(1), 1–14.
- Ibragimov, I.A., Khasminskii, R.Z. (1981). *Statistical Estimation: Asymptotic Theory*. Springer.
- Kutoyants, Yu.A. (1984). *Parameter Estimation for Stochastic Processes*. Heldermann.
- Kutoyants, Yu.A. (1998). *Statistical Inference for Spatial Poisson Processes*. Lecture Notes in Statist. 134, Springer.
- Ogata, Y. (1988). Statistical models for earthquake occurrences and residual analysis for point processes. *J. Amer. Statist. Assoc.* 83(401), 9–27.
- Rebolledo, R. (1980). Central limit theorems for local martingales. *Z. Wahrsch. Verw. Gebiete* 51, 269–286.